05 Thermodynamics-2

Limitation of first law of thermodynamics:

The first law of thermodynamics can evaluate the value of ΔU and ΔH (i.e. energy difference between two states), but cannot predict the feasibility of a process. Many processes can take place on their own and many others require an external force to carry out the process. The first law is clueless about why some processes happens on their own and why some others do not.

Importance of Second law:

What is the driving force for a process which takes system from one state to other on its own? Why such processes (**called spontaneous process**) take place at all? Answers to these questions are given by second law.

Spontaneous process:

A spontaneous process is the one which takes place on its own & can drive some other process as well. However, in some cases, initiation may be required.

Ex :

(i) Mixing of two gases.

(ii) Cooling of a hot object by transferring heat to surrounding.

(iii) Water flowing down the hill.

(iv) Rusting of iron.

(v) Burning of coal, coke, hydrocarbons.

(vi) Volcanic eruptions.

The synonyms of spontaneous processes are **natural processes** and irreversible processes.

It is a common observation that natural processes occur with finite rate. All spontaneous processes take place due to a finite driving force (which can be temperature difference, height difference, potential energy difference, etc.). Also, the change brought about by a spontaneous process cannot be restored without making some effort (which brings about permanent change in surrounding). Hence it is quite natural to call these processes as irreversible processes.

Ex :

When water flows from a dam, a part of the potential energy is converted into electrical energy and a part is converted into kinetic energy. The kinetic energy part is lost as heat energy. This loss of extra work is the permanent change in universe, because this heat energy lost cannot be completely converted back into P.E. of water.

Disorder:

The term 'disorder' is related to randomness. Any macroscopic property of a system is an average of large number of microscopic forces which are random in nature. Greater the randomness associated with microscopic forces, greater is the uncertainty about exact microscopic description of a macroscopic state and greater will be the disorder associated with the macroscopic state.

As a rule:

(i) Greater the number of molecules in which a given energy is distributed, greater is the disorder.

(ii) Greater the spread of given number of molecules with given energy in space, greater will be the disorder.

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Second law of thermodynamics:

The entropy of the universe increases in course of every spontaneous process. The entropy of system is a state function. The change in entropy of a system is given by -

$dS = \frac{dq_{rev}}{T}$ where ; dq_{rev} = heat exchanged by system in a very small step (reversibly).

Entropy:

Entropy is a measure of disorder. The entropy is a state function and a basic quantity directly defined by second law. In every spontaneous process, entropy of universe increases.

There are three types of entropy changes :

(i) ΔS_{System} : Entropy change of system

(ii) $\Delta S_{Surrounding}$: Entropy change of surrounding

(iii) ΔS_{Total} : Entropy change of universe

$\Delta S_{system} + \Delta S_{surrounding} = \Delta S_{universe} = \Delta S_{total} = \Delta S_{isolated}$

Normally, ΔS means entropy change of system.

Second law in terms of entropy of universe:

For any spontaneous process (irreversible process) -

$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surrounding} > 0$

During a reversible process -

$\Delta S_{universe} = 0$

Thus, during irreversible process, entropy of universe increases while during reversible process, entropy of universe remains constant.

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Physical significance of quantity $\frac{dq_{rev}}{T}$:

dq_{rev} = heat absorbed by the system, T = absolute temperature at which heat is absorbed. Suppose, similar quantity of heat is absorbed at two different temperatures T_1 and T_2 ($T_2 >> T_1$). When heat is supplied at high temperature, the % age increase in kinetic energy is less compared to heat supplied at low temperature. Hence, disorder will increase by greater value, when heat is supplied at lower temperature.

Determining '$\Delta S_{surr.}$' for irreversible processes :

The change in entropy is given by : $\frac{\text{heat absorbed by an entity}}{\text{Exact temperature at which heat is absorbed}}$

The irreversible processes are fast and uncontrolled. The exact temperature of system during exchange of heat cannot be known. Hence, reversible path is needed to evaluate the entropy of system. Along reversible path, the heat exchange and exact temperature both are known.

Surrounding is generally a big entity and heat absorbed by surrounding is comparatively very small as compared to its size. Hence, internal equilibrium of surrounding is not disturbed during exchange of heat.

$\therefore$ Actual heat exchange of system is reversible heat exchange of surroundings

$\therefore$ $(q_{rev})_{surr} = -q_{irr}$ [$\mathbf{q_{irr}}$ = Actual heat gained by system in a irreversible process.]

Hence, exact temperature and heat exchange both are known as far as surrounding is concerned even for an irreversible process.

$\therefore$ **For irreversible process :** $\Delta S_{surr} = \frac{-q_{irr}}{T}$

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	ΔS_{system}	$\Delta S_{surrounding}$
(i)	For reversible process : $\Delta S_{system} = \int_{T_1}^{T_2} \frac{dq_{rev}}{T}$	**For reversible process :** $\Delta S_{surr} = \int_{T_1}^{T_2} \frac{dq_{rev}}{T} = -\Delta S_{system}$
(ii)	For irreversible process : $\Delta S_{system} = \int_{T_1}^{T_2} \frac{dq_{rev}}{T}$ [Where; dq_{rev} = heat gained by system for imaginary reversible path between same states]	**For irreversible process :** $\Delta S_{surr} = -\frac{q_{irr}}{T}$
(iii)	Always evaluated along reversible path (for both reversible and irreversible process)	For reversible process, ΔS_{surr} is evaluated along reversible path, while for irreversible process, irreversible heat is used to calculate ΔS_{surr}.
(iv)	Since no actual process is reversible, ΔS_{system} is always evaluated by hypothetical path.	For hypothetical process, ΔS_{surr} is evaluated by hypothetical heat, while for real irreversible process, it is evaluated by heat exchanged in irreversible process.

Second law (in terms of engine):

(i) No cyclic engine is possible whose only effect is complete conversion of heat into work without making any change in surrounding on its own.

(ii) Heat cannot flow from cold body to hot body on its own without external intervention.

(iii) Complete conversion of heat (taken from a single source) into work is not possible in a cyclic process while complete conversion of work into heat is possible in a cyclic process. Thus, heat and work are not equivalent from the view point of II law.

All above statements can be easily justified by realising that heat is a disordered form of energy while work is an ordered form of energy.

Complete conversion of **[Heat ⟶ Work]** on its own without any external assistance means **[Disorder ⟶ Order]** on its own - which is impossible.

Entropy changes of an ideal gas:

From definition of entropy of system -

$$dS_{system} = \frac{dq_{rev}}{T} \qquad ...(i)$$

from first law, $dq_{rev.} = dU + PdV$

For a reversible change involving an ideal gas -

$$dq_{rev} = nC_{V,m}.dT + \frac{nRT}{V}.dV \qquad ...(ii)$$

$$\Rightarrow \int_1^2 \frac{dq_{rev}}{T} = \int_1^2 \frac{nC_{V,m}dT}{T} + \int_{V_1}^{V_2} nR\frac{dV}{V}$$

$$\Delta S_{system} = nC_{v,m}\ln\frac{T_2}{T_1} + nR\ell n\left(\frac{V_2}{V_1}\right) = nC_{p,m}\ \ln\frac{T_2}{T_1} - nR.\ln\frac{P_2}{P_1}$$

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Note :

This formula is applicable to all ideal gas process.

Ideal gas processes:

(i) For isothermal change :

$$\Delta S = \int_1^2 \frac{dq_{rev}}{T} = \frac{1}{T}\int_1^2 dq_{rev} \qquad (\because T = \text{constant})$$

$$\Rightarrow \Delta S = \frac{q_{rev}}{T} = nR\ell n\frac{V_2}{V_1} \qquad \left[\because q_{rev} = nRTln\left(\frac{V_2}{V_1}\right)\right]$$

(ii) For isochoric change :

$$\Delta S = \int \frac{dq_v}{T} = \int_1^2 \frac{nC_{V,m}dT}{T} = nC_{V,m}\int_1^2 \frac{dT}{T} \qquad [\because dq_v = nC_{V,m}.dT]$$

$$\Rightarrow \Delta S = nC_{V,m}\,\ell n\frac{T_2}{T_1}$$

(iii) For isobaric change :

$$\Delta S = \int \frac{dq_P}{T} = \int_1^2 \frac{nC_{P,m}.dT}{T} = nC_{P,m}\int_1^2 \frac{dT}{T} \qquad [\because dq_P = nC_{P,m}dT]$$

$$\Rightarrow \Delta S = nC_{P,m}\,\ell n\left(\frac{T_2}{T_1}\right)$$

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(iv) Reversible adiabatic process :

$q_{rev} = 0$

$$\Rightarrow \Delta S = \underbrace{nC_V ln\frac{T_2}{T_1}}_{\substack{\text{entropy change due to}\\ \text{change in temperature}}} + \underbrace{nRln\frac{V_2}{V_1}}_{\substack{\text{entropy change due to}\\ \text{Change in volume}}} = 0$$

(a) During reversible adiabatic expansion :

$T_2 < T_1$ and $V_2 > V_1$

$\Rightarrow$ Sign of two terms are opposite and magnitudes are same because for reversible adiabatic process -

$$nC_{V,m}\ln\frac{T_2}{T_1} = -nRln\frac{V_2}{V_1}$$

$\Rightarrow$ positive and negative term cancel out each other and hence, $\Delta S = 0$

Thus, decrease in entropy during expansion due to decrease in temperature is exactly cancelled out by increase in entropy due to increase in volume of system in reversible adiabatic expansion.

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(b) During reversible adiabatic compression:

$T_2 > T_1$ and $V_2 < V_1$

$$\Rightarrow \Delta S_{Total} = \underbrace{nC_{V,m}\ln\frac{T_2}{T_1}}_{\text{Positive}} + \underbrace{nR\ln\frac{V_2}{V_1}}_{\text{Negative}} = 0$$

Thus, increase in entropy due to increase in temperature is compensated by decrease in entropy due to decrease in volume of system in reversible adiabatic compression.

Hence, $\Delta S = 0$ for reversible adiabatic compression.

(v) Irreversible adiabatic process:

Since process is irreversible, $\Delta S_{Total} > 0$.

$\Rightarrow \Delta S_{system} + \Delta S_{surr} > 0$

$\because \Delta S_{surr} = 0$

$\because \Delta S_{system} > 0$

The physical interpretation is -

(a) During irreversible adiabatic compression, large quantity of heat is produced while decrease is volume is less.

Thus, increase in entropy due to rise in temperature exceeds the decrease in entropy due to decrease in volume. Hence net entropy change is positive.

(b) During irreversible adiabatic expansion, due to sudden decrease in external pressure, the system remains less challenged during expansion. Hence system do less work-resulting in lesser decrease in temperature.

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The increase in volume of system is relatively larger (because system was less challenged)

Thus, increase in entropy due to increase in volume exceeds decrease in entropy due to decrease in temperature. The net change in entropy is positive in both processes $\Delta S_{surr.} = 0$ ($\because$ no heat is exchanged with surrounding) and $\Delta S_{total} > 0$.

(vi) Free expansion of an ideal gas against vacuum:

Free expansion of an ideal gas is an example of irreversible adiabatic as well as irreversible isothermal process.

During free expansion, the $P_{ext.} = 0$. Hence, the ideal gas is not challenged at all during expansion. The kinetic energy of ideal gas remains constant. Hence, no temperature difference is created during expansion between system and surrounding.

$\because \Delta T = 0$, the heat absorbed during the process = 0

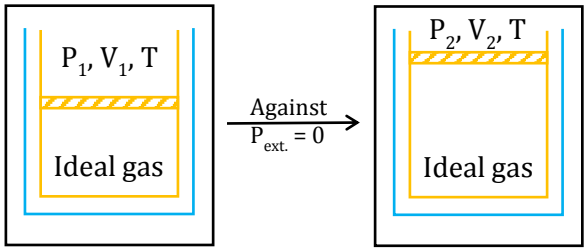
Since, $V_f > V_i$ and T = constant, $\Delta S_{system} > 0$

The free expansion is an example where $q_{actual} = 0$, still $\Delta S_{system} > 0$.

This is reasonable, because: $\Delta S_{system} = \int \frac{dq_{rev.}}{T}$

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This, example clearly shows that even if q_{actual} between two states is zero, this does not mean that $\int_1^2 \frac{dq_{rev.}}{T}$ between two states is zero. This outlines the importance of reversible path and $q_{rev.}$ in calculation of entropy of system.



$q_{irr} = 0$; $w_{irr} = 0$

$\Delta S_{system} = \int_1^2 \frac{dq_{rev.}}{T} = nR\ln\frac{V_2}{V_1}$

$\Delta S_{surr.} = \frac{-q_{irr}}{T} = 0$

$\Delta S_{total.} = nR\ln\frac{V_2}{V_1} > 0$

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ENTROPY CALCULATION

Process	$\Delta S_{Sys.}$	$\Delta S_{Surr.}$
Isothermal reversible	$\Delta S_{Sys.} = nR\ell n\frac{V_2}{V_1}$	$\Delta S_{Surr.} = -\Delta S_{Sys.}$
Isothermal irreversible	$\Delta S_{Sys.} = nR\ell n\frac{V_2}{V_1}$	$\Delta S_{Surr.} = \frac{-q_{Sys}}{T} = \frac{W_{Sys}}{T} = \frac{-P_{ext}(V_2 - V_1)}{T}$
Adiabatic reversible	$\Delta S_{Sys.} = 0$	$\Delta S_{Surr.} = 0$
Adiabatic irreversible	$\Delta S_{Sys.} = nC_{P,m}\ell n\frac{T_2}{T_1} + nR\ell n\frac{P_1}{P_2}$	$\Delta S_{Surr.} = 0$
Isochoric reversible	$\Delta S_{Sys.} = nC_{V,m}\ell n\frac{T_2}{T_1}$	$\Delta S_{Surr.} = -\Delta S_{Sys.}$
Isochoric irreversible	$\Delta S_{Sys.} = nC_{V,m}\ell n\frac{T_2}{T_1}$	$\Delta S_{Surr.} = \frac{-q_{sys}}{T_{surr}} = \frac{-nC_{V,m}\Delta T}{T_{final}}$

Illustration 1

Three moles of an ideal gas is expanded isothermally from 2L to 10L, at 27°C. Calculate ΔS_{sys} and ΔS_{surr} and $\Delta S_{univ.}$, if the process is performed.

(i) Reversibly

(ii) Irreversibly, against a constant external pressure of 1 atm

(iii) As free expansion

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Solution:

(i) $\Delta S_{univ.} = 0$, $\Delta S_{surr.} = -\Delta S_{sys.}$

and $\Delta S_{sys} = nR\ell n\frac{V_2}{V_1} = 3R.\ell n\frac{10}{2} = 40.14\ J/K$

(ii) $\Delta S_{sys} = 40.14\ J/K$

$\Delta S_{surr.} = -\frac{P_{ext.}(V_2 - V_1)}{T} = -\frac{1\times(10-2)\times 101.3}{300} = -2.7\ J/K$

and $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 37.44\ J/K$

(iii) $\Delta S_{sys} = 40.14\ J/K$

$\Delta S_{surr.} = 0$ as $P_{ext} = 0$

and $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 40.14\ J/K$

Illustration 2

Two moles of an ideal monoatomic gas is expanded adiabatically from 10L, 27°C to 80L. Calculate ΔS_{sys}, ΔS_{surr} and ΔS_{univ}, if the process is performed.

(i) Reversibly

(ii) Irreversibly, against a constant external pressure of 0.1 atm.

(iii) As free expansion

Solution:

$\Delta S_{surr} = 0$ for all

(i) $\Delta S_{sys} = \Delta S_{surr} = \Delta S_{univ} = 0$

(ii) $n\cdot C_{v,m}\cdot(T_2 - T_1) = -P_{ext}(V_2 - V_1)$

or $2\times\frac{3}{2}\times 0.0821\times(T_2 - 300) = -0.1\times(80-10) \Rightarrow T_2 = 271.58\ K$

Now, $\Delta S_{sys} = n\cdot C_{v,m}\cdot\ell n\frac{T_2}{T_1} + nR\cdot\ell n\frac{V_2}{V_1}$

$= 2\times\frac{3}{2}R\times\ell n\frac{271.58}{300}\times 2\times R\times\ell n\frac{80}{10} = 32.90\ J/K$

and $\Delta S_{univ} = \Delta S_{sys}$

(iii) $T_1 = T_2 \Rightarrow$ $\Delta S_{sys} = nR\cdot\ell n\frac{V_2}{V_1} = 2\times R\times\ell n\frac{80}{10} = 34.58\ J/K$

Entropy change of solids and liquids upon heating:

$\Delta S = \int\frac{dq_{rev}}{T} = \int\frac{mS.dT}{T} = mS.\ell n\frac{T_2}{T_1}$ [S = specific heat of solid/ liquid]

For molar change in entropy, m = M(g/mole) ; where M = molar mass.

Note, for solids and liquids ($S \cong C_P \cong C_V$)

Illustration 3

Two blocks of equal masses and heat capacity but at different temperatures are taken in an adiabatic vessel. Show that total change in entropy when both attain thermal equilibrium is always positive.

Solution:

Let the temperature of hot body be T_H, temperature of cold body be T_C & the final temperature be T_F.

Now, net heat exchanged by two bodies must be zero.

$\Rightarrow$ $mS(T_F - T_H) + mS(T_F - T_C) = 0$

$\Rightarrow$ $T_F = \frac{T_H + T_C}{2}$

Since both bodies are of finite mass, entropy of both bodies are evaluated by reversible paths.

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$$\Delta S_{Hot} = \int_{T_H}^{T_F} mS.\frac{dT}{T} = mS.\ell n\frac{T_F}{T_H}$$

$$\Delta S_{Cold} = \int_{T_C}^{T_F} mS.\frac{dT}{T} = mS.\ell n\frac{T_F}{T_C}$$

$$\Rightarrow \Delta S_{total} = (mS)\,\ell n\left(\frac{T_F^2}{T_H T_C}\right)$$

$$\Rightarrow T_H^2 + T_C^2 + 2T_H T_C - 4T_H T_C = (T_H - T_C)^2 > 0$$

$$\Rightarrow (T_H + T_C)^2 > 4T_H T_C$$

$$\Rightarrow \Delta S_{total} > 0$$

Reversible phase transitions:

Reversible phase transitions are always isothermal & isobaric.

$$\Delta S_{system,P,T} = \frac{q_{rev.}}{T} = \frac{\Delta H_{P,T}}{T_{Transition}}$$

$$\Delta S_{surr} = \frac{-\Delta H_{P,T}}{T_{Transition}}$$

$\Delta S_{total} = 0$

Thus, $\Delta S_{vaporisation}$ at boiling point $= \dfrac{\Delta H_{vaporisation}}{T_{vaporisation}}$

ΔS_{fusion} at melting point $= \dfrac{\Delta H_{fusion}}{T_{fusion}}$

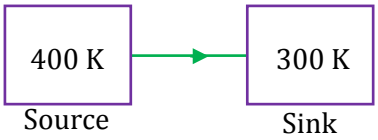
$\Delta S_{sublimation}$ at sublimation point $= \dfrac{\Delta H_{sublimation}}{T_{sublimation}}$

$\Delta S_{allotropic\ modification} = \dfrac{\Delta H_{allotropic\ modification}}{T_{transition}}$

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Illustration 4

100 kJ heat is transferred from a larger heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat :



Find ΔS_{source}, ΔS_{sink} and ΔS_{total}. Comment on spontaneity of process.

Solution:

(i) $\Delta S_{source} = \dfrac{\text{heat absorbed by source}}{\text{Temperature of source}}$

$$\Delta S_{source} = \frac{-100kJ}{400} = -250J/K$$

(ii) $\Delta S_{sink} = \dfrac{\text{heat absorbed by sink}}{\text{Temperature of sink}}$

$$= \frac{+100000J}{300} = 333.33J/K$$

(iii) $\Delta S_{Total} = \Delta S_{source} + \Delta S_{sink} = +83.33\ J/K$

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Illustration 5

Trouton's rule states that the ratio of the molar heat of vaporisation of a liquid to its normal boiling point is approximately the same for all liquids, $\frac{\Delta H_{vap}}{T_{bp}} = 88\ JK^{-1}\ mol^{-1} = \Delta S_{vap.}$

Which of the following liquids do not follow Trouton's rule : NH_3, C_6H_6, CCl_4, $CHCl_3$, Hg. Explain.

Solution:

NH_3, due to H-bonding : $\Delta S_{vap} > 88$ J/K-mol

Third law of thermodynamics

"At absolute zero, the entropy of a perfectly crystalline substance is zero", which means that at absolute zero, every crystalline solid is in a state of perfect order and its entropy should be zero.

By virtue of the third law, the absolute value of entropy (unlike absolute value of enthalpy) for any pure substance can be calculated at room temperature.

$$S_T - S_{0K} = \int_0^T \frac{q_{rev}}{T}$$

Since $S_{0K} = 0$

$$S_T = \int_0^T \frac{q_{rev}}{T}$$

The variation of entropy of a substance with temperature is given by graph :

At 1, entropy of substance is zero.

- **from 1-2 ;** The temperature of solid increases upon heating -

  $\Rightarrow S_2 - S_1 = \int \frac{C_{P,solid}}{T} dT$ ...(i)

- **from 2-3 ;** The temperature remains constant during phase transition

  $\Rightarrow S_3 - S_2 = \Delta S_{fusion} = \frac{\Delta H_{fusion}}{T_f}$ ...(ii)

- **from 3-4 ;** The temperature of liquid increases

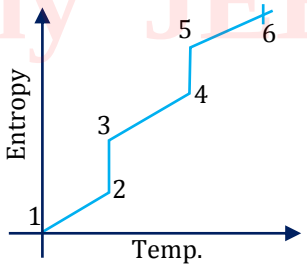
  $\Rightarrow S_4 - S_3 = \int_{T_3}^{T_4} \frac{C_{P,liquid}.dT}{T}$ ...(iii)

- **from 4-5 ;** Reversible vaporisation takes place at constant temperature -

  $\Rightarrow S_5 - S_4 = \Delta S_{vap.} = \frac{\Delta H_{vap.}}{T_b}$ ...(iv)

- **from 5-6 ;** The gas is heated -

  $\Rightarrow S_6 - S_5 = \int_{T_5}^{T_6} \frac{C_{P,gas}.dT}{T}$ ...(v)



Change in entropy of a pure substance with temperature at constant pressure.

$\Delta S_{1-6} = (S_2 - S_1) + (S_3 - S_2) + (S_4 - S_3) + (S_5 - S_4) + (S_6 - S_5) = S_6 - S_1$ $[\because S_1 = 0]$

$\because$ S_6 = absolute entropy at 6.

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$$= \int_{T_1}^{T_2} \frac{C_{P,Solid}dT}{T} + \frac{\Delta H_{fusion}}{T_f} + \int_{T_3}^{T_4} \frac{C_{P,liquid}dT}{T} + \frac{\Delta H_{vap}}{T_b} + \int_{T_5}^{T_6} \frac{C_{P,gas}dT}{T}$$

- **Comment :** Absolute entropy of a substance can be calculated using third law by reversible path.

Standard absolute molar entropy :

It is the entropy in one mole of substance at standard temperature at 1 bar.

It is calculated assuming that substance is heated isobarically and reversibly at 1 bar from 0 K to standard temperature.

Comparing absolute entropies of substance :

(i) The absolute entropies of substance follow the order,

$S(g) >> S(l) > S(s)$

(ii) Entropy increases when solid and liquid is dissolved into solvent,

e.g.	$CH_3OH(l)$	$S° = 127$ J/K-mole
	$CH_3OH(aq)$	$S° = 132.3$ J/K-mole
	NaCl(s)	$S° = 72.4$ J/K-mole
	NaCl(aq.)	$S° = 115.4$ J/K-mole

(iii) The entropy of dissolved gas is less than pure gas.

e.g.	HCl(g)	$S° = 186.7$ J/K-mole
	HCl(aq)	$S° = 55.2$ J/K-mole

(iv) Entropy rises with increasing mass, (if other factors are similar).

e.g.	$F_2(g)$	$S° = 203$ J/K-mole
	$Cl_2(g)$	$S° = 223$ J/K-mole
	$Br_2(g)$	$S° = 245$ J/K-mole

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(v) Entropy is lower in covalently bonded solids, with strong directional bonds, than in solids with partial metallic character.

e.g.	C(diamond)	$S° = 2.44$ J/K-mole
	C(graphite)	$S° = 5.69$ J/K-mole
	Sn(gray)	$S° = 44.8$ J/K-mole
	Sn(white)	$S° = 51.5$ J/K-mole

(vi) Entropy increases with increasing softness and with weakness of bonds between atoms.

e.g.	C(diamond)	Be(s)	$S_1O_2(s)$	Pb(s)	Hg(l)
	2.44	9.54	41.8	64.9	77.4
	diamond	hard metal	quartz	soft metal	liquid

(vii) Entropy increases with chemical complexity

Sub.	NaCl	$MgCl_2$	$AlCl_3$
	72.4	89.5	167

For $CuSO_4.nH_2O$

n = 0	n = 1	n = 3	n = 5
$S° = 113$	150	225	305

Note :

Above data is given in SI unit.

Entropy change ($\Delta_r S$) of a chemical reaction :

The change in entropy when reactants turn into products according to a balanced chemical reaction with specified physical state of each reactant and product is called entropy of reaction.

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- **Consider a reaction :**
  $aA(m) + bB(n) \xrightarrow{T,P} cC(o) + dD(p)$
  where m, n, o, p are physical states at temperature T and pressure P.
  $\Delta S_r = \sum S$ (products) – $\sum S$ (reactants)
  $\Delta_r S = cS_C + dS_D - aS_A - bS_B$
  S_A, S_B, S_C and S_D are absolute molar entropies at temperature T and pressure P **(from third law)**
  So, $\Delta_r S$ is evaluated from table of data of third law entropies of various substances.

$\Delta S_{Surrounding}$ for chemical reaction :

$\Delta_r H$ = q_p = heat absorbed by the system during chemical reaction.

$\Rightarrow$ **$-\Delta_r H$** = $-\ q_p\ =\ q_{surr.}$ = heat absorbed by the surrounding.

Now,

$$\boxed{\Delta S_{Surr} = \frac{q_{surr}}{T} = -\frac{\Delta_r H}{T}}$$

If $\Delta_r H$ = –ive (exothermic) heat is lost by the system and gained by the surrounding. Hence, ΔS_{surr} increases.
If $\Delta_r H$ = +ive (endothermic process), entropy of surrounding decreases.

$\because\quad \Delta S_{surr} = -\frac{\Delta_r H}{T}$ = –ive

- We cannot comment on entropy change of system from knowledge of $\Delta_r H$. The entropy changes of system ($\Delta_r S$) is estimated by third law.

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Driving force of a spontaneous chemical process:

The driving force of a spontaneous chemical process is increase in entropy of universe.
In simple terms, negative value of $\Delta_r H$ and positive value of $\Delta_r S$ favours the chemical process.
If $\Delta_r H$ is positive, the enthalpy factor don't support the forward process. Under these conditions, $\Delta_r S$ should be sufficiently positive to drive the reaction in forward direction. If $\Delta_r H$ is positive and $\Delta_r S$ is negative, under a given situation, the process become non-spontaneous because both factors tend to decrease the entropy of the universe.

Prediction the sign of $\Delta_r S$ by inspecting a balanced chemical reaction

(i) If more no. of gaseous moles are present on product side, $\Delta_r S$ will be positive (since gas is more disordered than solid or liquid).

(ii) If solid $\longrightarrow$ liquid
solid $\longrightarrow$ gas
liquid $\longrightarrow$ gas
then, $\Delta_r S$ = +ive.

(iii) If a molecule is undergoing cyclisation, it's rotational modes of motion are replaced by vibrational modes of motion.
Since quantum energy separation of rotational motion is closely spaced compared to vibrational energy state, rotational energy is more disordered. Hence, $\Delta_r S$ = –ive.

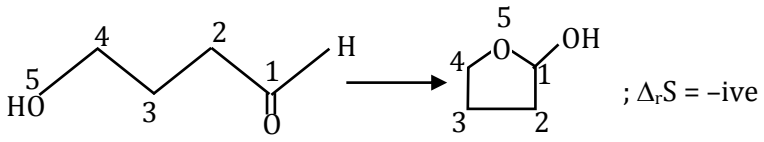
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- **Example :**

$HCOOH(g) \longrightarrow H_2(g) + CO_2(g)$; Δn_g = +ive $\Rightarrow \Delta_r S$ = +ive

$H_2O(\ell) \longrightarrow H_2O(g)$; $\Delta_r S$ = +ive

$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(g)$; Δn_g = +ive $\Rightarrow \Delta_r S$ = +ive



; $\Delta_r S$ = –ive

Gibb's function:

Gibb's function 'G' is defined as G = H–TS.

Since 'G' is a combination of state variables (a compound state variable), it is a state function. The dimension of Gibb's function is the dimension of energy. 'G' is an extensive quantity.

Gibb's function provides a very useful criteria for spontaneity of a process taking place at constant temperature and pressure (isobaric, isothermal process).

The entropy of universe is also a very useful criteria of spontaneity, but it is based on properties of system as well as surrounding.

The Gibb's function provides criteria for spontaneity based only on the properties of system.

$\Delta G = \Delta H - \Delta(TS)$

$\Delta G = \Delta H - T\Delta S$ (Isothermal process)

Illustration 6

The thermodynamic stability of a substance is dependent upon the value of Gibb's function. Explain the stability of solid at low temperature and stability of gas at very high temperature.

Solution:

Any stable state tries to minimise its enthalpy while maximising its entropy.

For solids - $G_{solid} = H_{solid} - TS_{solid}$ (at certain T & P)

For liquids - $G_{liquid} = H_{liquid} - TS_{liquid}$ (at certain T & P)

For gas - $G_{gas} = H_{gas} - TS_{gas}$ (at certain T & P)

At any given T and P, $H_{solid} < H_{liquid} < H_{gas}$

and $S_{solid} < S_{liquid} < S_{gas}$

At very low temperature, TS term contributes very little towards Gibb's function. Hence -

$G_{solid} < G_{liquid} < G_{gas}$ (at very low temperature).

This explains why solids are most stable at low temperature.

At very high temperature, TS term is very important. Since S_{gas} is large, for gas TS term becomes very large at large temperature. Hence -

$\Rightarrow$ $G_{solid} > G_{liquid} > G_{gas}$ (at very high temperature). Thus, gaseous state is most stable at high temperature. Hence, the low enthalpies of solid are responsible for stability of solid at low temperature, while high entropy of gas is responsible for stability of gas at high temperature.

$\Delta G_{T,P}$ and $\Delta S_{universe}$

Decrease in Gibb's function at constant temperature and pressure is related to ΔS_{total} (total entropy change of system and surrounding).

$-\Delta G = T(\Delta S_{total})$

At constant P, T

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if $\Delta G < 0 \Rightarrow$ spontaneous process
if $\Delta G > 0 \Rightarrow$ non-spontaneous process
if $\Delta G = 0 \Rightarrow$ reversible process

Table for spontaneity of chemical process and sign of $\Delta_r H$, $\Delta_r S$ and dependence on temperature.

Sign of ΔH	Sign of ΔS	Comment	Example	$\Delta H°_{298}$	$\Delta S°_{298}$
–	+	Spontaneous at all temperature	$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ $C(s) + O_2(g) \rightarrow CO_2(g)$	–185 –394	14.1 3
–	–	Spontaneous at low temperature	$H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(\ell)$ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	–44 –198	–119 –187
+	+	Spontaneous at high temperature	$NH_4Cl(s) \rightarrow NH_3(g) + HCl(g)$ $N_2(g) + O_2(g) \rightarrow 2NO(g)$	176 180	284 25
+	–	Non-spontaneous at all temperature	$3O_2 \rightarrow 2O_3$ $2H_2O(\ell) + O_2(g) \rightarrow 2H_2O_2(\ell)$	286 196	–137 –126

State of equilibrium:

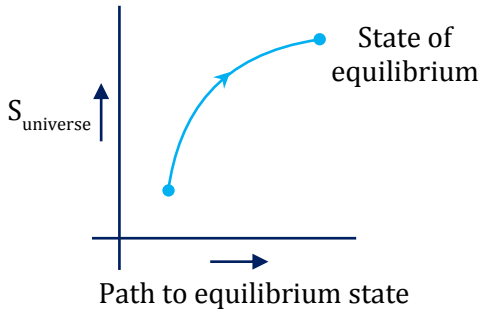
A system at equilibrium is a stable system. Hence, its state variables become constant (do not vary with time) at equilibrium.

If a system is in non-equilibrium state, it moves spontaneously in the direction of equilibrium (because the equilibrium state is stable). During the spontaneous movement towards equilibrium, the entropy of universe increases.

As the system gets closer to equilibrium state, the driving force to attain equilibrium become small and so does the increase in disorder of universe. At equilibrium, the driving force to move ahead finishes and entropy of the universe cannot increase further.

In nutshell, when a system attains equilibrium, it's capacity to increase the disorder of universe is lost.

- At equilibrium, the entropy of universe is maximum.



Calculate ΔG for various processes:

(i) Ideal gas subjected to isothermal compression or expansion:

For isothermal process:

$\Delta G = \Delta H - T\Delta S$

For ideal gas, $\Delta H = nC_{P,m}\Delta T = 0$ (As $\Delta T = 0$, for isothermal change)

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& $\Delta S = nR \ln \frac{V_2}{V_1}$ (for isothermal change)

$\therefore \quad \Delta G = -nRT \ln \frac{V_2}{V_1}$

Hence, during isothermal expansion Gibb's function decreases while during isothermal compression Gibb's function increases.

(ii) Isothermal and isobaric phase transitions:

At transition temperature, the stability (or Gibb's functions) of two phases are equal. Hence, there is no driving force for conversion of one phase into other. This means phase transitions are examples of reversible processes at transition temperature.

Now, for reversible, isothermal and isobaric change, $\Delta G = 0$

$\Rightarrow$ $\Delta G = 0$, for phase transitions at transition temperature.

Thus, $\Delta G_{vap} = 0$ at boiling point :

$\Rightarrow \Delta H_{vap} = T_b \Delta S_{vap}$

$\Rightarrow T_b = \frac{\Delta H_{vap}}{\Delta S_{vap}}$

Also, $\Delta G_{fusion} = 0$ at melting point :

$\Rightarrow \Delta H_{fusion} = T_f \Delta S_{fusion}$

$T_f = \frac{\Delta H_{fusion}}{\Delta S_{fusion}}$

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Illustration 7

The enthalpy of vaporization of a liquid is 30 kJ mol^{-1} and entropy of vaporization is 75J mol^{-1} K^{-1}. The boiling point of the liquid at 1 atm is:

(A) 250 K (B) 400 K (C) 450 K (D) 600 K

Ans. (B)

Solution:

$$T = \frac{\Delta H_{vap}}{\Delta S_{vap}} = \frac{30 \times 10^3}{75} = 400K$$

Illustration 8

At what minimum temperature would a given reaction become spontaneous if $\Delta H = +119$ kJ and $\Delta S = +263$ J/K?

(A) 452 K (B) 2210 K (C) 382 K (D) 2.21 K

Ans. (A)

Solution:

$$T = \frac{\Delta H}{\Delta S} = \frac{119 \times 10^3}{263} = 452.47K$$

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(iii) $\Delta_r G$ for a chemical reaction

Consider a reaction: $aA + bB \xrightarrow{T,P} cC + dD$

The change in Gibb's function for a chemical reaction is change in ΔG when reactants react according to a given balanced chemical reaction with specified physical state of each reactant and product at given temperature and pressure.

$\Delta_r G = cG_C + dG_D - aG_A - bG_B$ ...(i)

Where G_A, G_B, G_C and G_D are molar Gibb's function of A,B,C and D respectively at given T and P.

also,

$$\left.\begin{aligned} G_A &= H_A - TS_A \\ G_B &= H_B - TS_B \\ G_C &= H_C - TS_C \\ G_D &= H_D - TS_D \end{aligned}\right\} \quad ...(ii)$$

Substituting (ii) in (i)

$\Delta_r G = (cH_C + dH_D - aH_A - bH_B) - T(cS_C + dS_D - aS_A - bS_B)$

$\Rightarrow \quad \boxed{\Delta_r G = \Delta_r H - T\Delta_r S}$

$(\Delta H_r)_{T,P}$	$(\Delta S_r)_{T,P}$	(ΔG_r)	Remarks
– ve	+ve	Always –ve	Reaction is spontaneous
+ ve	– ve	Always +ve	Reaction non-spontaneous
+ ve	+ ve	At low temperature, ΔG = + ve At high temperature, ΔG = – ve	Non-spontaneous Spontaneous
– ve	– ve	At low temperature, – ve	Spontaneous
– ve	– ve	At high temperature, + ve	Non-spontaneous

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(iv) Gibb's functions of pure substances :

Consider a pure substance having a Gibb's function as follows :

$G = H - TS = U + PV - TS$

Let, the pressure and temperature are varied by infinitesimally small value :

$dG = dU + PdV + VdP - TdS - SdT$

As, $dU = dq - PdV$ and $TdS = dq$ [Assume only PV- work and reversible process.]

$dG = VdP - SdT$

(a) If pressure is varied keeping temperature constant -

$$dG = VdP \Rightarrow \left(\frac{dG}{dP}\right)_T = V$$

The physical significance of above relation is when pressure over a substance is increased, it's Gibb's function increases & the increase is proportional to molar volume. Since molar volume of gas, liquid and solid are in order, as follows :

$V_{Gas} > V_{Liquid} > V_{Solid}$

Hence, Gibb's function of gases increases most rapidly on increasing pressure.

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That's why, gas phase becomes unstable at high pressure.

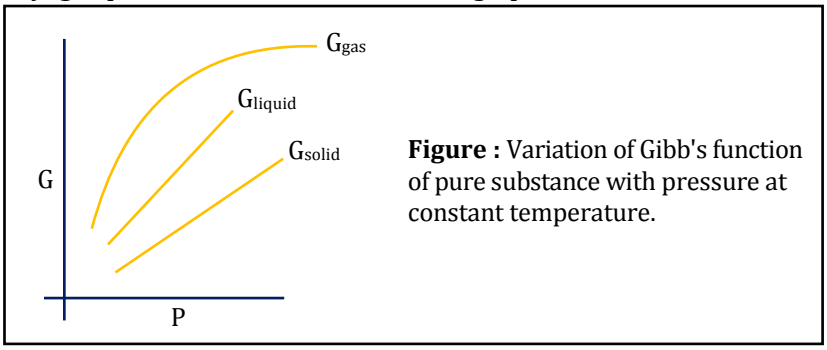


Figure : Variation of Gibb's function of pure substance with pressure at constant temperature.

(b) If temperature is changed keeping pressure constant -

$$dG = -SdT \Rightarrow \left(\frac{dG}{dT}\right)_P = -S$$

Since entropy of pure substances are always positive, the slope of G vs T curve is negative. Also, since,

$\mathbf{S_{gas} > S_{liquid} > S_{solid}}$

$\Rightarrow$ the slope is more negative for gas.

Thus, on increasing temperature, the Gibb's function of pure substances decreases, but this decrement is maximum for gaseous phase. Thus, gas phase acquire stability on increasing temperature.

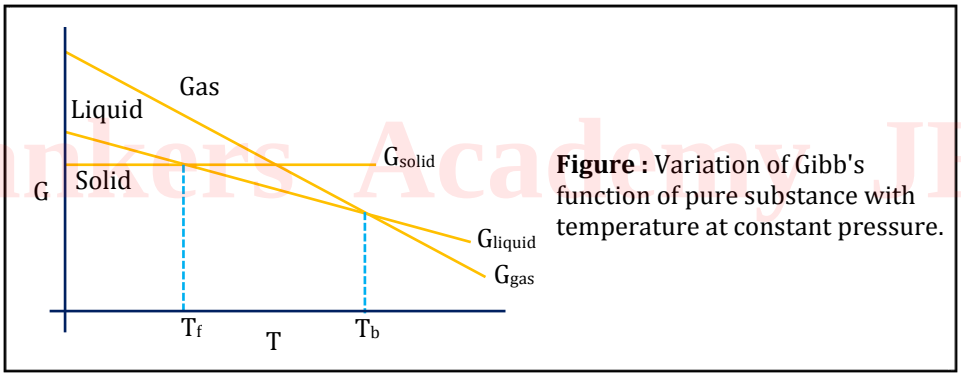


Figure : Variation of Gibb's function of pure substance with temperature at constant pressure.

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Illustration 9

Calculate ΔG_m for the following process:

(i) $H_2O\ (\ell, 100°C, 1\ atm) \longrightarrow H_2O\ (g, 100°C, 1\ atm)$

(ii) $H_2O\ (\ell, 100°C, 1\ atm) \longrightarrow H_2O\ (g, 100°C, 0.5\ atm)$

Solution:

(i) $\Delta G = 0$ (Equilibrium condition)

(ii) $H_2O\ (\ell, 100°C, 1atm) \xrightarrow{1} H_2O\ (g, 100°C, 1\ atm) \xrightarrow{2} H_2O\ (g, 100°C, 0.5\ atm)$

$\Delta G_1 = 0$

$$\Delta G_2 = nRT \cdot \ell n\frac{P_2}{P_1} = 1 \times R \times 373 \times \ell n\frac{0.5}{1} = -2149.5\ J$$

$\therefore \quad \Delta G = \Delta G_1 + \Delta G_2 = -2149.5\ J$

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Application of Gibb's energy (as measure of non-PV work):

For a pure substance

$dG = d(H - TS) = dH - TdS - SdT$

also, $dH = dU + PdV + VdP$

also, $dU = dq - PdV - w_{non\text{-}PV}$

($w_{non\text{-}PV}$ = non-PV work done by system)

$\Rightarrow dG = dq - PdV - w_{non\text{-}PV} + PdV + VdP - TdS - SdT$

for a reversible change at constant T and P.

dq = TdS ; and SdT and VdP term vanishes.

$\Rightarrow dG = TdS - w_{non\text{-}PV} - TdS$

$(dG)_{T,P} = - w_{non\text{-}PV}$

$\Rightarrow \boxed{-(dG)_{T,P} = w_{non-PV}}$

Thus, decrease in Gibb's function at constant temperature and pressure is equal to maximum non-PV work obtainable from system.

Chemical equilibrium:

The state of chemical equilibrium is characterised by constant value of temperature, pressure and composition of a closed system. The state of equilibrium represents most stable state acquired by chemical system under given conditions.

The characteristics of state of equilibrium are :

(i) The equilibrium is attained in closed system.

(ii) At equilibrium, the chemical system is at minimum value of Gibb's function possible under given conditions.

(iii) A state of equilibrium is attained spontaneously.

(iv) At equilibrium, the rate of change of Gibb's function with progress of reaction under given conditions becomes zero $\Rightarrow \Delta_r G = 0$.

(v) At state of equilibrium, the total Gibb's function of products becomes equal to total Gibb's function of reactants.

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Equilibrium constant:

Gas phase reactions :

Consider a reaction : **$aA(g) + bB(g) \longrightarrow cC(g) + dD(g)$**

Now molar Gibb's functions of A, B, C and D are given by :

$G_A = G°_A + RT\ln\left(\frac{P_A}{P_0}\right)$ $\qquad$ $G_B = G°_B + RT\ln\left(\frac{P_B}{P_0}\right)$

$G_C = G_C° + RT\ln\left(\frac{P_C}{P_0}\right)$ $\qquad$ $G_D = G°_D + RT\ln\left(\frac{P_D}{P_0}\right)$

Now, $\Delta_r G = cG_C + dG_D - aG_A - bG_B$

Substituting value of G_A, G_B, G_C and G_D in above equation :

$$\Delta_r G = (cG_C° + dG°_D - bG°_B - aG°_A) + RT\ln\left[\frac{\left(\frac{P_C}{P_0}\right)^c\left(\frac{P_D}{P_0}\right)^d}{\left(\frac{P_A}{P_0}\right)^a\left(\frac{P_B}{P_0}\right)^b}\right]$$

$\Delta_r G = \Delta_r G° + RT\ln Q$

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Where, $Q = \left(\frac{P_C^c . P_D^d}{P_A^a . P_B^b}\right)\left(\frac{1}{P_0}\right)^{(c+d-a-b)}$

At equilibrium, $\Delta_r G = 0$ and $Q = K_P = \left(\frac{P_C^c . P_D^d}{P_A^a . P_B^b}\right)\left(\frac{1}{P_0}\right)^{\Delta n_g}$

$$\Delta_r G^\circ = -RT\ln K_P$$

Note:

Q contains partial pressures at that moment but K_P contain partial pressures at equilibrium.

Relationship between, ΔG° and equilibrium constant:

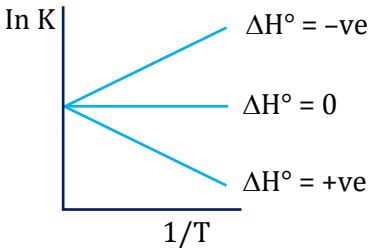
$\Delta G^\circ = -RT \ln Keq$

$\Delta H^\circ - T\Delta S^\circ = -RT \ln Keq$

$$\ln K_{eq} = \frac{-\Delta_r H^\circ}{RT} + \frac{\Delta_r S^\circ}{R}$$

$$\ln K_1 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_1}$$

$$\ln K_2 = \frac{\Delta_r S^\circ}{R} - \frac{\Delta_r H^\circ}{RT_2}$$



[where K_1 & K_2 are values of K_p at temperature T_1K & T_2K respectively]

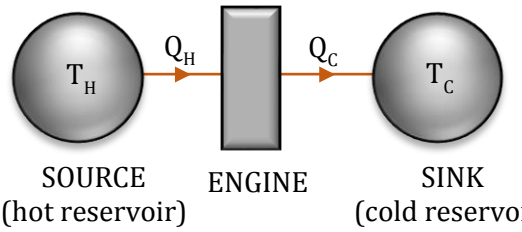
$$\ln\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$ (Van't Hoff's equation)

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Heat engine:

A heat engine is a device which can work continuously and cyclically without any external help absorbing heat from a source and rejecting a part of heat into sink. The net heat absorbed by engine (heat absorbed from source – heat given out to sink) is converted into work.

⇒ Heat engine must work in cycles, otherwise it cannot work continuously.



Q_H = Heat gained by engine per cycle

Q_C = Heat rejected by engine per cycle

Efficiency of engine : $\eta = \frac{Q_H - Q_C}{Q_H} = \frac{w_{by}}{Q_H}$

W_{by} = Work done by engine per cycle.

During operation of heat engine, heat is transferred from source (high temperature reservoir) to sink (low temperature reservoir) thus degrading the quality of energy. Note that net energy of universe remains conserved, but its quality is degraded (i.e. more disordered energy) when heat engine runs spontaneously.

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Carnot engine:

A carnot engine (a hypothetical engine) runs in reversible manner. Thus, during expansion (work producing process) maximum work is done. However, during compression (work requiring process) minimum work need to be done. Thus, no engine can be more efficient than carnot engine.

The carnot engine works in following steps. (Assume ideal gas as working material).

(i) Reversible isothermal expansion $(P_1, V_1, T_2 \rightarrow P_2, V_2, T_2)$

(ii) Reversible adiabatic expansion $(P_2, V_2, T_2 \rightarrow P_3, V_3, T_1)$

(iii) Reversible isothermal compression $(P_3, V_3, T_1 \rightarrow P_4, V_4, T_1)$

(iv) Reversible adiabatic compression $(P_3, V_3, T_1 \rightarrow P_1, V_4, T_2)$

Now :

$$q_{NET} = -w_{NET} \quad ...(i)$$

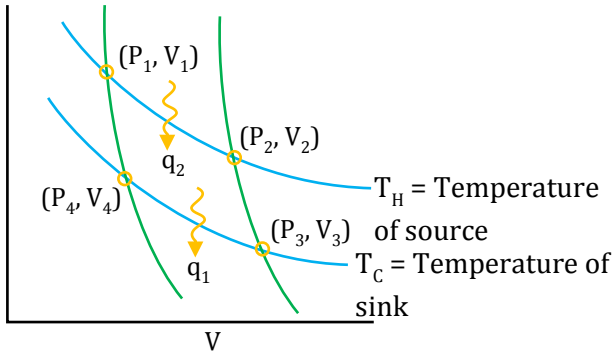
$$w_{NET} = w_{12} + w_{23} + w_{34} + w_{41} \quad ...(ii)$$

$$w_{12} = -nRT_H \ln\left(\frac{V_2}{V_1}\right) \quad ...(iii)$$

$$w_{23} = nC_V(T_C - T_H) \quad ...(iv)$$

$$w_{34} = -nRT_C \ln\left(\frac{V_4}{V_3}\right) \quad ...(v)$$

$$w_{41} = -nC_V(T_H - T_C) \quad ...(vi)$$



Also, since process 2 – 3 and 4 – 1 are reversible adiabatic process.

$$T_H V_2^{\gamma-1} = T_C V_3^{\gamma-1} \quad ...(vii) \quad \text{for process 2 – 3}$$

$$T_C V_1^{\gamma-1} = T_H V_4^{\gamma-1} \quad ...(viii) \quad \text{for process 1 – 4}$$

from above equations

$$\left(\frac{V_2}{V_1}\right) = \left(\frac{V_3}{V_4}\right) \quad ...(ix)$$

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Substituting the value of $\left(\frac{V_4}{V_3}\right)$ in eq.(v) and adding w in individual step.

$$w_{NET} = -nRT_H \ln\left(\frac{V_2}{V_1}\right) + nC_V(T_C - T_H) - nRT_C \ln\left(\frac{V_4}{V_3}\right) - nC_V(T_C - T_H)$$

$$w_{NET} = -nRT_H \ln\left(\frac{V_2}{V_1}\right) - nRT_C \ln\left(\frac{V_1}{V_2}\right)$$

$$w_{NET} = -nR\left[T_C - T_H\right]\ln\left(\frac{V_2}{V_1}\right)$$

$\Rightarrow$ Efficiency $\eta = -\frac{w_{NET}}{q_2}$

$$\eta = -\frac{nR\ln\left(\frac{V_2}{V_1}\right)[T_H - T_C]}{nRT_H \ln\left(\frac{V_2}{V_1}\right)}$$

$\therefore \quad \eta = \frac{T_H - T_C}{T_H}$

Thus, efficiency of Carnot engine depends only upon temperature of source and sink.

Note :

If working substance is changed (eg. ideal gas by real gas) efficiency of Carnot engine remains unaffected.

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